

1(a)	work done per unit mass	B1
	work done moving mass from infinity (to the point)	B1
1(b)(i)	potential is zero at infinity	B1
	work is done by (two) masses in moving them closer together or work is done on (two) masses in moving them apart	B1
1(b)(ii)	magnitude of potential shown as 4ϕ	B1
	potential negative and shown as a multiple of $-\phi$ [potential = -4ϕ if fully correct]	B1
1(b)(iii)	field strength at X: $\phi/4R$	A1
	field strength at Y: $4\phi/R$	A1
	potential energy at X: $-M\phi$	A1
	potential energy at Y: $-8M\phi$	A1